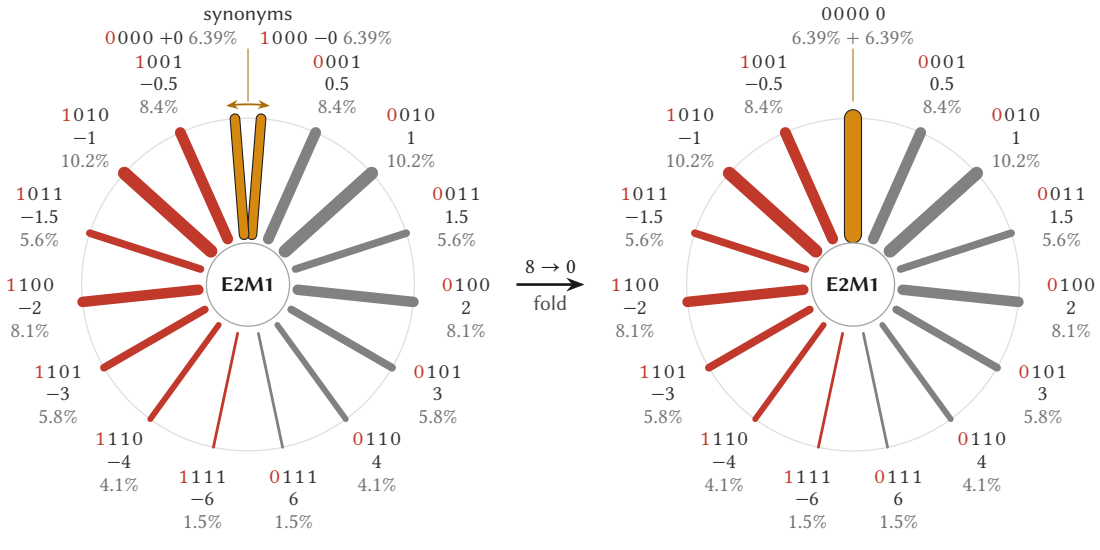
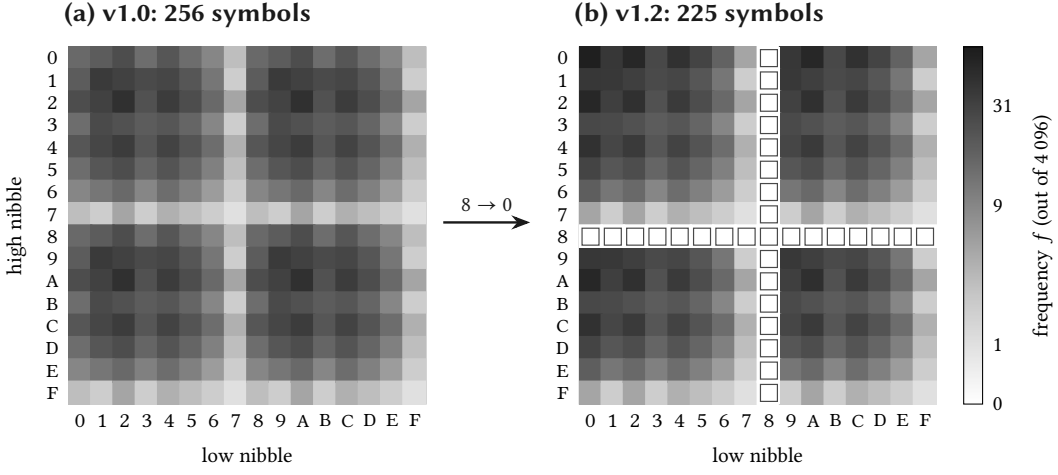


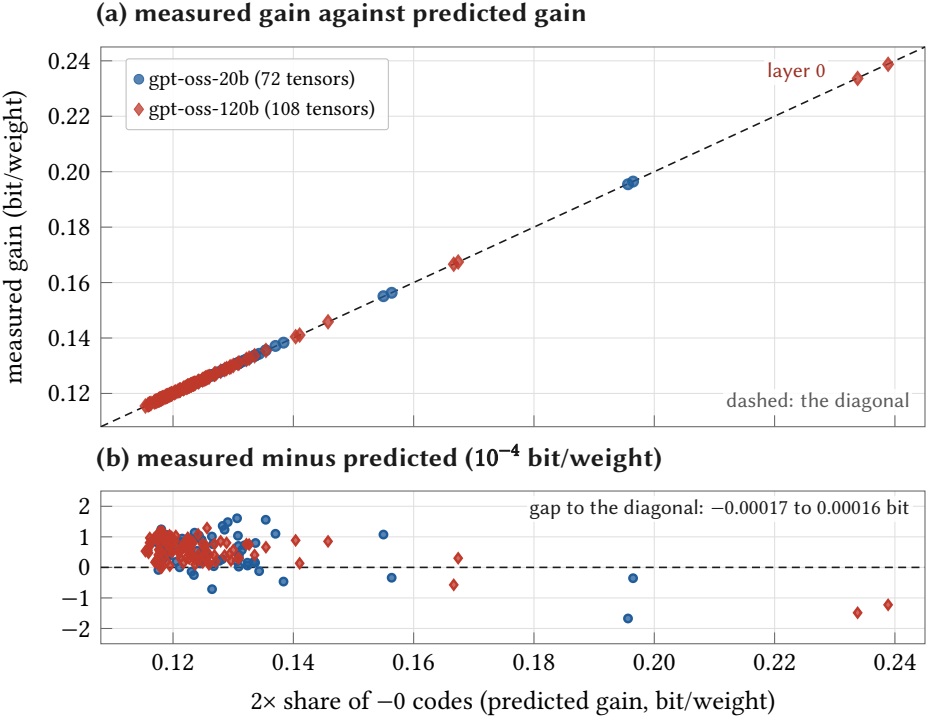
(a) before: 16 codes, 15 values

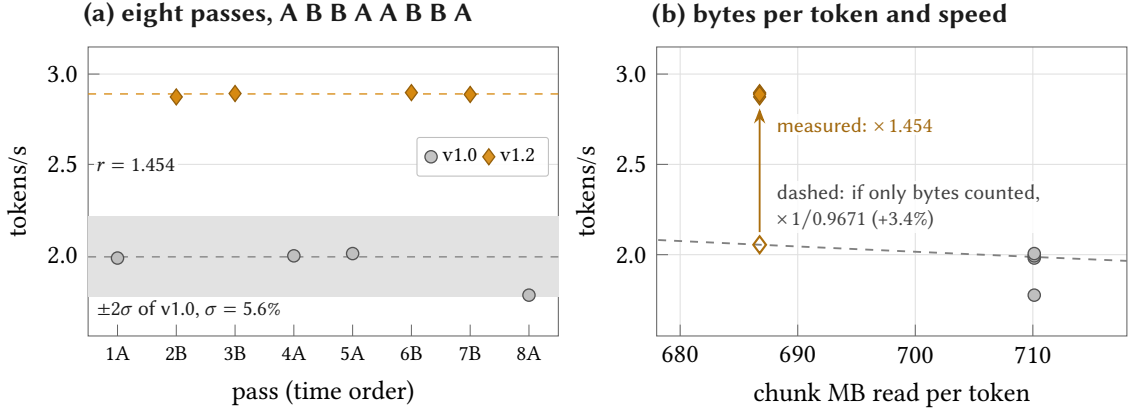
(b) after: 15 codes, 15 values



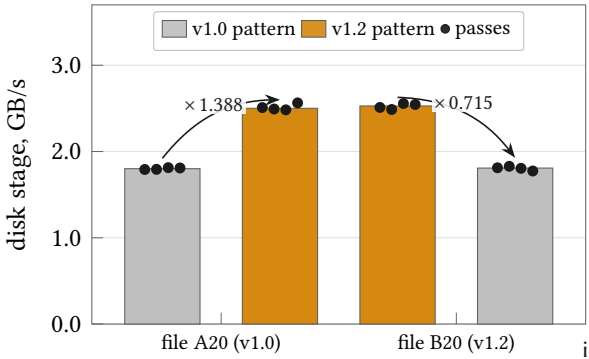
the sign of a zero is a fair coin: $H(S \mid V = 0) = 0.9999999967$ bit
$P(+0) = 6.3920\%$, $P(-0) = 6.3929\%$ (the same 72 tensors of the 20b)

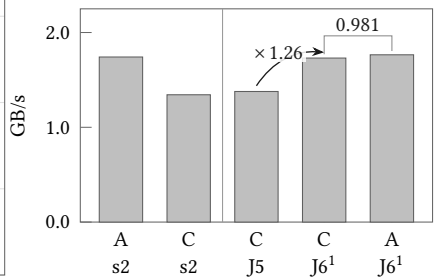

(a) v1.0: 256 symbols
(b) v1.2: 225 symbols
high nibble
low nibble
8 → 0
frequency f (out of 4 096)
31
9
1
0


(a) measured gain against predicted gain
measured gain (bit/weight)
0.24
0.22
0.20
0.18
0.16
0.14
0.12
gpt-oss-20b (72 tensors)
gpt-oss-120b (108 tensors)
layer 0
dashed: the diagonal
(b) measured minus predicted (10−4 bit/weight)
2
1
0
−1
−2
gap to the diagonal: −0.00017 to 0.00016 bit
0.12
0.14
0.16
0.18
0.20
0.22
0.24
2× share of −0 codes (predicted gain, bit/weight)


(a) eight passes, A B B A A B B A
3.0
2.5
2.0
tokens/s
r = 1.454
v1.0
v1.2
±2σ of v1.0, σ = 5.6%
1A
2B
3B
4A
5A
6B
7B
8A
pass (time order)
(b) bytes per token and speed
3.0
2.5
2.0
tokens/s
measured: × 1.454
dashed: if only bytes counted,
× 1/0.9671 (+3.4%)
680
690
700
710
chunk MB read per token

(a) same file, other read pattern (20b)





inset (120b, disk stage): A120 and C120, two v1.0 files equal to the bit; [1] one-extent chunks only

(b) the speed decomposed (120b, three files in one session)

